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Metabolomic and Transcriptomic Profiling of Two Closely Related Species within the Genus *Oldenlandia*

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The genus *Oldenlandia* encompasses numerous medicinal plants with significant pharmacological potential, yet their tissue-specific metabolic pathways and transcriptional profiles remain understudied. This study combined transcriptomic and metabolomic analyses to compare the differences in five tissues (roots, stems, leaves, flowers, and fruits) between two closely related *Oldenlandia* species, *Oldenlandia diffusa* and *Oldenlandia corymbosa*. Using a widely targeted metabolomics approach based on UPLC-MS/MS, we monitored 1,343 metabolites across 60 samples from the two species. Transcriptome sequencing generated 272.12 Gb and 257.03 Gb of clean data for *Oldenlandia diffusa* and *Oldenlandia corymbosa*, respectively, along with 37,644 and 20,825 annotated genes. Comparative analysis revealed numerous differentially expressed genes, whose diverse functions were elucidated through GO enrichment. This comprehensive dataset provides a valuable resource for further investigating the phenotypic traits and metabolic mechanisms underlying the pharmacological activities of these medicinally important *Oldenlandia* species.

Background & Summary

The genus *Oldenlandia* (Rubiaceae) comprises over 400 species native to tropical and subtropical regions, such as India, Myanmar, Sri Lanka, Cambodia, Indonesia, and tropical Africa[1]. *Oldenlandia diffusa* (Willd.) Roxb, a medicinal species of *Oldenlandia* genus, is known for its abilities to clear heat, detoxify, promote diuresis, reduce swelling, and enhance blood circulation to alleviate pain. Similarly, *Oldenlandia corymbosa* L., another species in the same genus, exhibits comparable pharmacological properties[2]. Both species are extensively utilized in traditional medicine for treating inflammatory diseases, liver ailments, and various types of cancer[2–4]. They contain diverse bioactive secondary metabolites, including flavonoids, phenolic acids, alkaloids, and terpenoids[2]. As closely related species, they share similar morphological traits, with minor distinctions in their stems, leaves, flowers, and fruits. *Oldenlandia diffusa* features either flattened or cylindrical stems, long and slender leaves, solitary or opposite flowers in leaf axils with short, thick pedicels, and oblate-spherical capsules. The stem of *Oldenlandia corymbosa* is quadrangular, with axillary flowers typically arranged in corymbs of 2 to 5, featuring long, slender pedicels and spherical capsules[5,6] (Fig. 1).

Due to their similar efficacy and properties, the mixed use of *Oldenlandia diffusa* and *Oldenlandia corymbosa* is prevalent in the market[5,7]. The ambiguity in the application of this medicinal material, combined with the currently limited comparative studies, makes it necessary for us to emphasize the need to use systematic multi-omics analysis to address the identification of specific characteristics of medicinal plants and optimize their clinical applications. The development of metabolomics and transcriptomics has enabled us to have a deeper understanding of the secondary metabolic processes in plants. The integration of transcriptomic and metabolomics data provides a new perspective for understanding the phenotypic traits, metabolic pathways and gene regulation of *Oldenlandia* plants[8–10]. Metabolites often accumulate in plants and are transported across different tissue parts, and then are translocated to storage sites within the plants or organs, resulting in a metabolic landscape with differential distribution across various plant tissues[11]. Currently, research on plants

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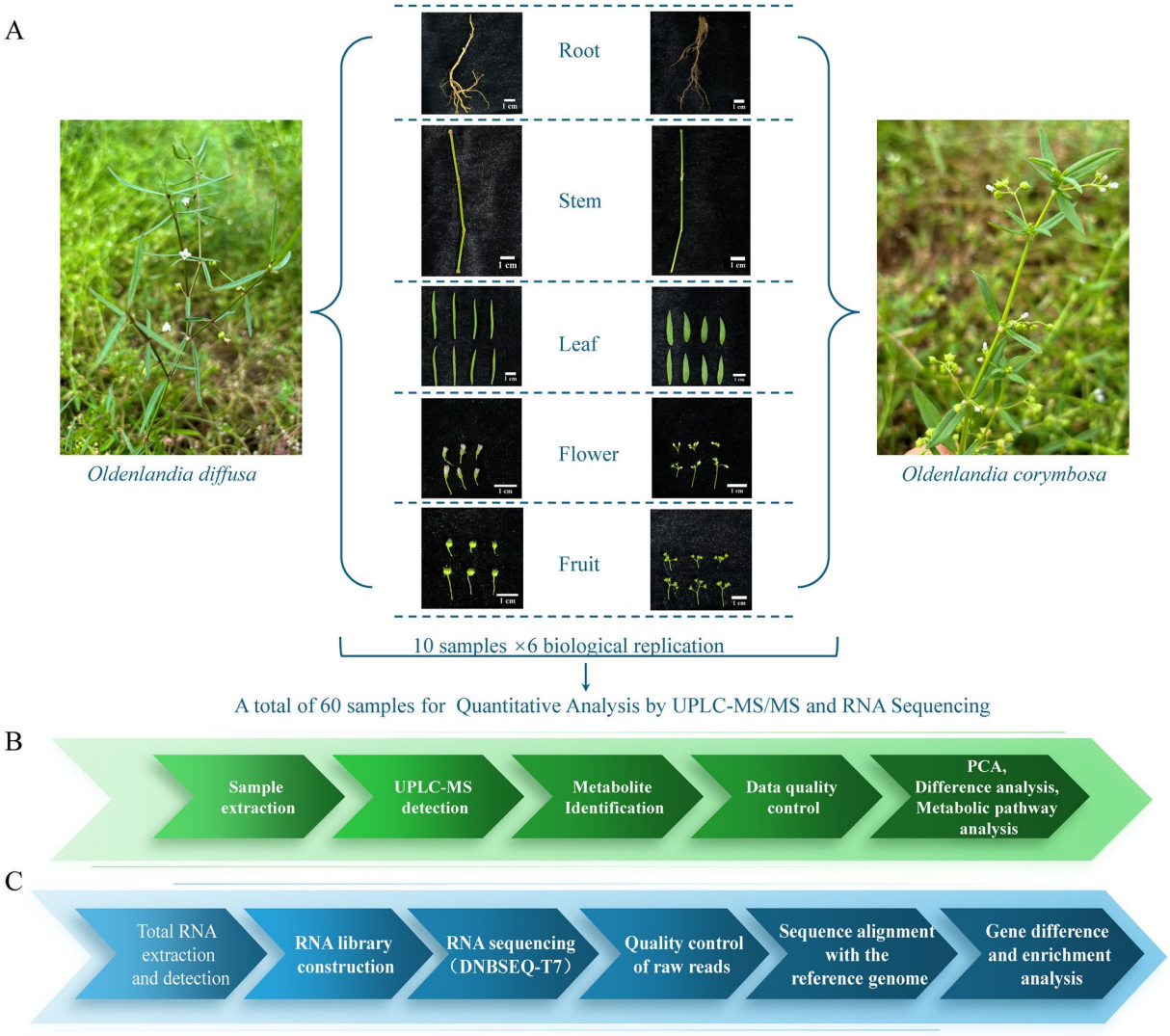


Fig. 1 Sampling of *Oldenlandia diffusa* and *Oldenlandia corymbosa* (**A**). Select 5 tissue parts (Root, Stem, Leaf, Flower, Fruit) of the two plants during their flowering period (**B,C**) Simplified metabolomics and transcriptomic sequencing analysis process.

of the *Oldenlandia* genus mainly focuses on the activity analysis of single metabolites or extracts[3,12,13], these studies typically homogenize plant tissues and organs, overlooking the tissue-specific metabolic characteristics. Moreover, metabolites, acting as a bridge between genes and phenotypes, can elucidate the molecular mechanisms underlying relevant phenotypes and promote the study of gene functions[14]. Integrating metabolomics and transcriptomics at the cellular or tissue level has proven valuable in investigating plant secondary metabolic pathways and traits. By analyzing the coordinated changes in gene expression and metabolite levels, candidate genes are pinpointed, subsequently validated through techniques like gene editing, overexpression/silencing, and enzymatic assays. Following this research framework, key enzymes such as squalene epoxidase, triterpene synthase, epoxide hydrolase, cytochrome P450, and UDP-glucosyltransferase have been identified and functionally characterized in the mogroside V biosynthetic pathway of Siraitia grosvenorii, shedding light on the genetic basis of sweetness traits[15]; In Taxus biosynthesis, utilizing a chromosome-level genome, key enzymes like CYP725A were identified through an integrated transcriptional and metabolic expression network. Their pathway roles were validated via enzymatic and cellular system interventions, elucidating the mechanistic link from genes to medicinal phenotypes[16]. The recently reported chromosome-level genome assemblies of *Oldenlandia diffusa* and *Oldenlandia corymbosa* provide important references for transcriptome sequencing[17,18]. Previous studies have identified two key genes for ursolic acid biosynthesis in *Oldenlandia corymbosa*[18]. However, due to the limited research on the metabolic mechanisms of *Oldenlandia* plants, the biosynthetic patterns of their key active metabolites remain unclear. Moreover, comparative analyses between the closely related species *Oldenlandia diffusa* and *Oldenlandia corymbosa* could contribute to uncovering the drivers of interspecific metabolic diversity, but such studies are scarce.

In this study, we collected samples from five distinct tissues—root, stem, leaf, flower, and fruit—of both *Oldenlandia diffusa* and *Oldenlandia corymbose* (Fig. 1A). Utilizing widely targeted metabolomics based on

UPLC-QTOF-MS/MS, we monitored secondary metabolites and compared interspecific differential metabolites within the same tissue types. Transcriptomic data were generated using second-generation RNA sequencing based on the DNBSEQ-T7 platform, yielding 272.12 Gb and 257.03 Gb of clean data for the respective species. Subsequent data quantification and differential gene analysis were conducted. Quality control for both metabolomic and transcriptomic datasets involved principal component analysis and hierarchical clustering. These datasets serve as a crucial reference for future investigations into plant phenotypes and metabolic mechanisms.

Methods

Collection of plant sample materials. *Oldenlandia diffusa* and *Oldenlandia corymbosa* were obtained from a traditional Chinese medicine plantation in Xinzhou City, Hubei Province (Piaoer Mountain, Jiujie Street, Xinzhou: N30° 49' 40.42", E114° 56' 43.54"). These two species were cultivated together and shared similar flowering periods. Samples were collected during this period from the same nursery. Each tissue sample was derived from six biological replicates, with each replicate consisting of a pooled mixture of corresponding tissues from six individual plants. Roots, stems, leaves, flowers, and fruits were dissected and collected separately (Fig. 1A). The roots were excised at the stem-root junction, followed by the sequential separation of leaves, flowers, and fruits. Following photography, all tissue samples were immediately placed in 50 mL conical tubes, rapidly frozen in liquid nitrogen, and stored at −80 °C. From each biological replicate, two aliquots were allocated for metabolomics and transcriptomics analyses, respectively.

Metabolite analysis using a widely targeted metabolomics approach based on UPLC-MS/MS. Widely targeted metabolomics using multiple reaction monitoring (MRM) is a sensitive and precise technique for quantifying plant metabolites[19]. The workflow from sample extraction to data analysis was depicted in Fig. 1B. Biological samples were vacuum freeze-dried using a lyophilizer (Scientz-100F) and ground to a fine powder (30 Hz, 1.5 min) with a grinder (MM 400, Retsch). A 50 mg aliquot of the powdered sample was weighed using an electronic balance (MS105DM) and mixed with 1200 μL of pre-cooled (−20 °C) 70% methanolic aqueous internal standard extract (adjusted proportionally if less than 50 mg of sample was used). The mixture was vortexed for 30 seconds every 30 minutes, repeated three times. After centrifugation at 12,000 rpm for 3 minutes, the supernatant was filtered through a 0.22 μm microporous membrane and stored in an injection vial for UPLC-MS/MS analysis. The extracts were analyzed using a UPLC-ESI-MS/MS system (UPLC, ExionLC™ AD, https://sciex.com.cn/) and tandem mass spectrometry (https://sciex.com.cn/). The UPLC analytical conditions include an Agilent SB-C18 column (1.8 μm, 2.1 mm × 100 mm) with a moving phase of solvent A (pure water with 0.1% formic acid) and solvent B (acetonitrile with 0.1% formic acid). Sample measurements employed a gradient program starting at 95% A and 5% B. A linear gradient to 5% A and 95% B was achieved in 9 minutes, maintained for 1 minute. The composition was then adjusted to 95% A and 5% B within 1.1 minutes and held for 2.9 minutes. The flow velocity was 0.35 mL/min; the column oven was set to 40 °C; and the injection volume was 2 μL. The effluent was connected to an ESI-triple quadrupole-linear ion trap (QTRAP)-MS. ESI source parameters as follows: source temperature 500 °C; ion spray voltage (IS) 550 V (positive ion mode) / −450 V (negative ion mode); ion source gases I (GSI), II (GSII), and curtain gas (CUR) set at 50, 60, and 25 psi, respectively; with high collision-activated dissociation (CAD). QQQ scans were conducted as MRM experiments with nitrogen as the collision gas set to medium. Declustering potential (DP) and collision energy (CE) for individual MRM transitions were optimized, with specific MRM transitions monitored according to the metabolites eluted during each period.

Principal component analysis (PCA) was performed using the prcomp function in the R4.4.2 statistical software (www.r-project.org). The data was unit variance scaled prior to the unsupervised PCA. For the two-group analysis, differential metabolites were monitored based on a variable importance in projection (VIP) score greater than 1 and an absolute fold change (|FC|) of at least 2. The data was log-transformed (log_2) and mean-centered before conducting the orthogonal partial least squares discriminant analysis (OPLS-DA). To avoid overfitting, a permutation test with 200 permutations was performed. The monitored metabolites were annotated using the KEGG compound database[20] (http://www.kegg.jp/kegg/compound/), and the annotated metabolites were then mapped to the KEGG Pathway database[21] (http://www.kegg.jp/kegg/pathway.html). Pathways with significantly regulated metabolites were then subjected to metabolite set enrichment analysis (MSEA), and their significance was determined by the hypergeometric test p-values.

Transcriptome sequencing and analysis of gene expression levels. The process of RNA sequencing and transcriptome data analysis is outlined in Fig. 1C. Total RNA was extracted from various plant tissues, including Roots, Stems, Leaves, Flowers, and Fruits, using the TRIZOL reagent kit (Magen, Guangzhou, China). RNA concentration and total quantity were assessed using the Qubit 4.0 fluorometer. RNA integrity was evaluated with the Agilent 2100 Bioanalyzer, while sample purity was determined via the NanoDrop spectrophotometer. Eukaryotic mRNA was enriched from total RNA using the Hieff NGS® mRNA Isolation Master Kit (Yeasen Cat#12603) with oligo(dT)-conjugated magnetic beads. The purified mRNA was fragmented in the presence of fragmentation buffer, followed by first-strand cDNA synthesis primed with random hexamers. Second-strand cDNA was then synthesized to generate double-stranded cDNA, which underwent end repair, poly(A) tail addition, and adapter ligation. The resulting products were purified and size-selected using Hieff NGS® DNA Selection Beads. Subsequently, PCR amplification was performed to construct the sequencing library. Following quantification via Qubit, libraries were sequenced on the DNBSEQ-T7 platform, and raw sequencing reads were stored in FASTQ format[22]. The raw reads were filtered using the Fastp software (https://github.com/OpenGene/fastp), and the quality of the cleaned data was assessed using the FastQC software (http://www.bioinformatics.babraham.ac.uk/projects/fastqc). The reference genome information for *Oldenlandia corymbosa* was obtained

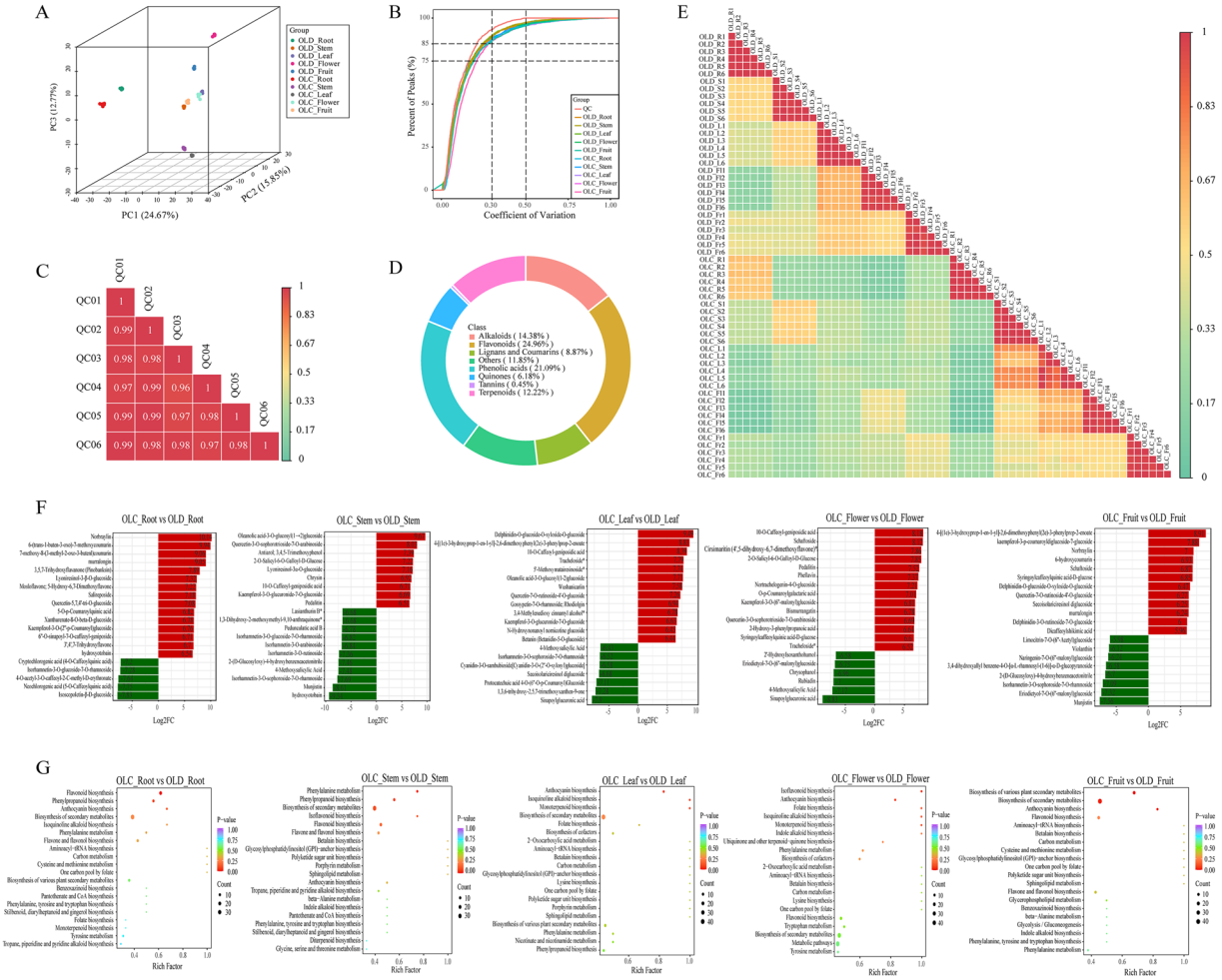


Fig. 2 Correlation analysis, differential analysis, and enrichment analysis of metabolomic data (**A**). Principal component analysis (PCA) plot of metabolites (**B**). CV distribution curves of samples in each group (**C**). Correlation plot among quality control samples(QC) (**D**). Circular plot of metabolite category composition (**E**). Heatmap analysis of inter - group correlations of all metabolite samples (**F**). Differential metabolite analysis of the same tissue parts of two plants (top 20) (**G**). KEGG enrichment analysis of the same tissue parts of two plants (top 20). Note: OLC and OLD represent samples from *Oldenlandia diffusa* and *Oldenlandia corymbosa* respectively. R, S, L, Fl, and Fr represent roots, stems, leaves, flowers, and fruits respectively. -1, -2, -3, -4, -5, -6 represent six biological replicates.

from the NCBI website[18], while a self-assembled reference genome was used for *Oldenlandia diffusa*. The quality-controlled RNA sequence was then compared to the reference genome using the hisat2 software[23]. The quality-controlled RNA sequences were compared to the reference genome. StringTie software[24] was then used to assemble the transcription of each sample, and the resulting transcripts were merged into a comprehensive set. The gffcompare software was employed to compare this transcript set to the reference annotation file, enabling the identification of both novel genes and new transcripts of known genes. To quantify gene expression levels, the FPKM (fragments per kilobase of exon per million reads mapped) metric was calculated. Differential expression analysis was performed using the DESeq2 software[25], with a threshold of $\log_2|\text{FoldChange}| \geq 2$ and $\text{Padj} \leq 0.05$. Finally, the clusterProfiler software[26] was utilized to conduct GO enrichment analysis, considering significant results as those with padj (FDR) < 0.05.

Identification of differential expression metabolites (DEMs) between *Oldenlandia diffusa* and *Oldenlandia corymbosa*. To investigate the differential metabolites between two *Oldenlandia* species, we conducted five paired comparisons of the same tissue parts: OLC_Root vs OLD_Root, OLC_Stem vs OLD_Stem, OLC_Leaf vs OLD_Leaf, OLC_Flower vs OLD_Flower, and OLC_Fruit vs OLD_Fruit. This allowed us to monitor the differential metabolites between the roots, stems, leaves, flowers, and fruits of the two plants, respectively. Qualitative and quantitative analysis revealed the top 20 metabolites with the highest fold change in each comparison group, which were visualized using bar charts (Fig. 2F). The KEGG enrichment analysis of the five tissues showed that the differentially expressed metabolites (DEMs) were significantly enriched in the biosynthetic pathways of various secondary metabolites, such as flavonoid biosynthesis, phenylpropanoid biosynthesis,

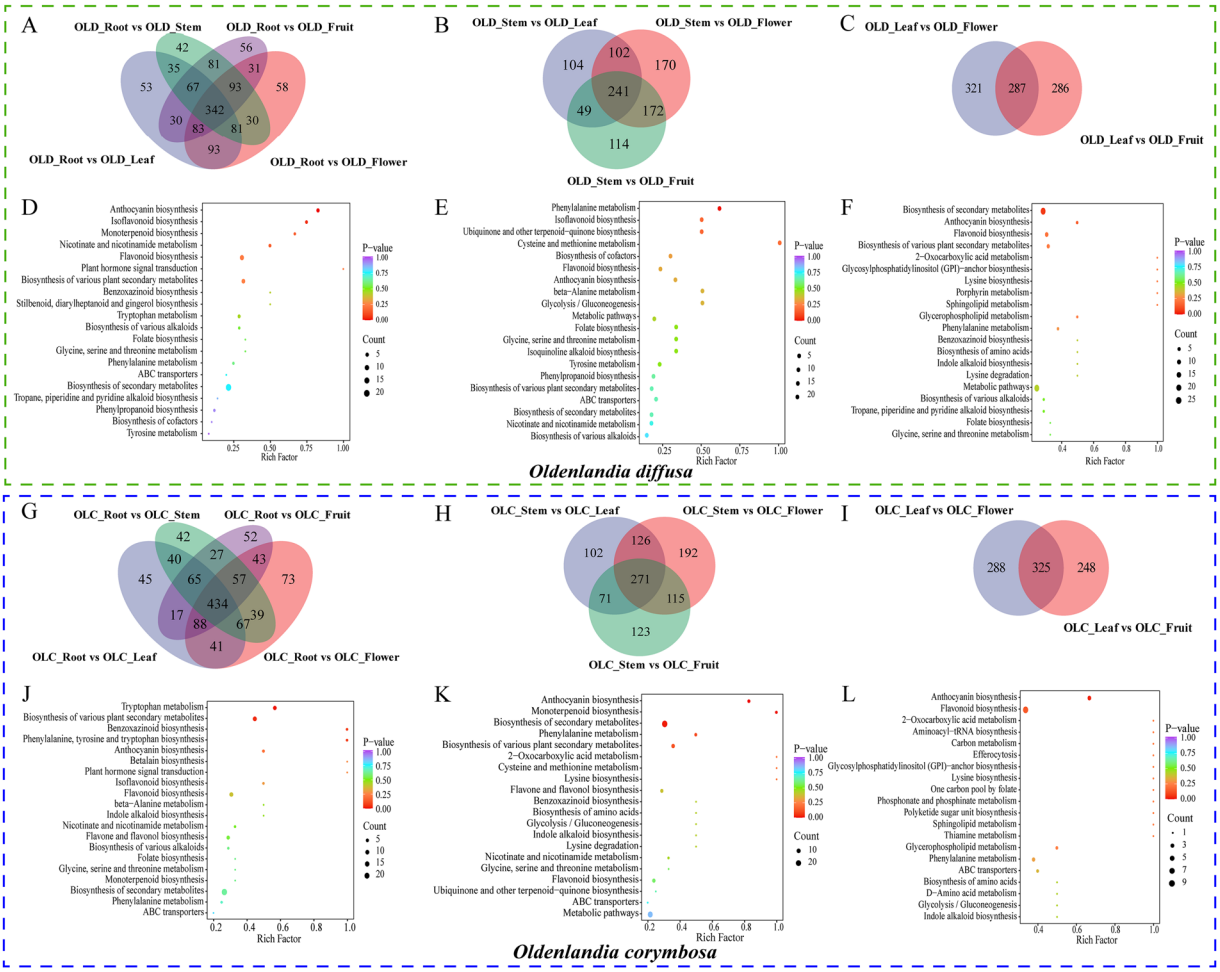


Fig. 3 Metabolome enrichment analysis within species (**A**). OLD_Root vs OLD_Stem, OLD_Root vs OLD_Leaf, OLD_Root vs OLD_Flower, OLD_Root vs OLD_Fruit (**B**). OLD_Stem vs OLD_Leaf, OLD_Stem vs OLD_Flower, OLD_Stem vs OLD_Fruit, and (**C**) OLD_Leaf vs OLD_Flower and OLD_Leaf vs OLD_Fruit compare Venn diagrams sharing DEMs between combinations (**D**–**F**) KEGG analysis of the shared DEMs among the comparison combinations of A-C indications (**G**); OLC_Root vs OLC_Stem, OLC_Root vs OLC_Leaf, OLC_Root vs OLC_Flower, OLC_Root vs OLC_Fruit (**H**) OLC_Stem vs OLC_Leaf, OLC_Stem vs OLC_Flower, OLC_Stem vs OLC_Fruit, and I: OLC_Leaf vs OLC_Flower and OLC_Leaf vs OLC_Fruit compare Venn diagrams sharing DEMs between combinations (**J**–**L**) KEGG analysis of the shared DEMs among the comparison combinations of G-I indications.

phenylalanine metabolism, anthocyanin biosynthesis, and isoflavonoid biosynthesis. This indicates that the different tissues are involved in A variety of metabolic activities (Fig. 2G). Our metabolomic dataset uncovered the distinct metabolic profiles in various tissues of *Oldenlandia diffusa* and *Oldenlandia corymbosa*, providing a crucial foundation for assessing species- and tissue-specific functions.

Identification of differential metabolites (DEMs) among different tissues of *Oldenlandia diffusa* and *Oldenlandia corymbosa*. Venn diagrams were constructed to illustrate the shared differentially expressed metabolites (DEMs) across various comparison groups in two species (Fig. 3A–C, G–I). The first Venn diagram compared DEMs between root tissues and above-ground tissues (stem, leaf, flower, and fruit) to monitor metabolite differences between underground and aerial parts. In *Oldenlandia diffusa*, a total of 342 commonly detected DEMs were monitored (Fig. 3A), which were predominantly enriched in anthocyanin biosynthesis, isoflavonoid biosynthesis, monoterpenoid biosynthesis, and nicotinate and nicotinamide metabolism (Fig. 3D). In contrast, *Oldenlandia corymbosa* exhibited 434 shared DEMs (Fig. 3G), primarily associated with the biosynthesis of various plant secondary metabolites, phenylalanine metabolism, tropane, piperidine and pyridine alkaloid biosynthesis, and tryptophan metabolism (Fig. 3J).The second Venn diagram analyzed the overlapping DEMs among stem and other above-ground tissues through pairwise comparisons: stem vs. leaf, stem vs. flower, and stem vs. fruit. In *Oldenlandia diffusa*, 241 shared DEMs were detected (Fig. 3B), mainly involved in phenylpropanoid biosynthesis, ubiquinone and other terpenoid-quinone biosynthesis, and isoflavonoid biosynthesis (Fig. 3E). In *Oldenlandia corymbosa*, 271 shared DEMs were monitored (Fig. 3H), which were enriched in pathways including biosynthesis of secondary metabolites, monoterpenoid biosynthesis, and anthocyanin biosynthesis (Fig. 3K).

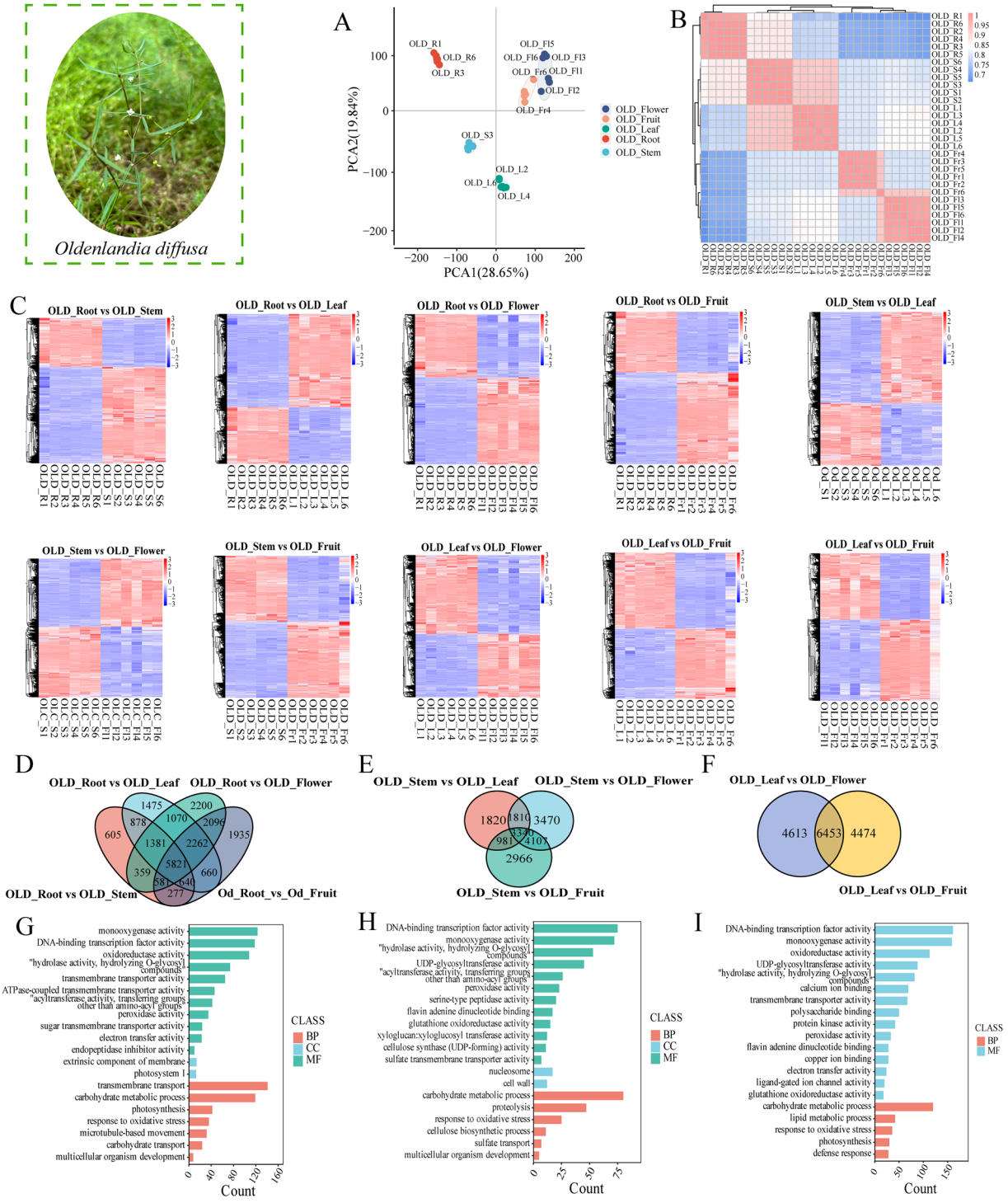


Fig. 4 Transcriptome data analysis of *Oldenlandia diffusa*, including expression profiles and DEG analysis for each comparative combination (**A**). Principal component analysis (PCA) plot of transcriptome samples (**B**). Heat map of gene associations between transcriptome samples (**C**). DEG expression pattern for each paired combination. Up-regulated DEGs are highlighted in red; down-regulated DEGs are shown in blue. DEG with padj < 0.05 and $|FC| > 2$ screening (**D**). OLD_Root vs OLD_Stem, OLD_Root vs OLD_Leaf, OLD_Root vs OLD_Flower, OLD_Root vs OLD_Fruit (**E**). OLD_Stem vs OLD_Leaf, OLD_Stem vs OLD_Flower, OLD_Stem vs OLD_Fruit, and (**F**). OLD_Leaf vs OLD_Flower and OLD_Leaf vs OLD_Fruit compare Venn diagrams sharing DEG between combinations (**G**). GO analysis of shared DEG between OLD_Root vs OLD_Stem, OLD_Root vs OLD_Leaf, OLD_Root vs OLD_Flower, OLD_Root vs OLD_Fruit comparison combinations (top 20 GO entries) (**H**). GO analysis of shared DEG between OLD_Stem vs OLD_Leaf, OLD_Stem vs OLD_Flower, OLD_Stem vs OLD_Fruit comparison combinations (top 20 GO entries) (**I**). GO analysis of shared DEG between OLD_Leaf vs OLD_Flower, and OLD_Leaf vs OLD_Fruit comparison combinations (top 20 GO entries). Note: OLD represents samples from *Oldenlandia diffusa*, R, S, L, Fl, Fr represent Roots, Stems, Leaves, Flowers and Fruits respectively. -1, -2, -3, -4, -5, -6 represent six biological repeat.

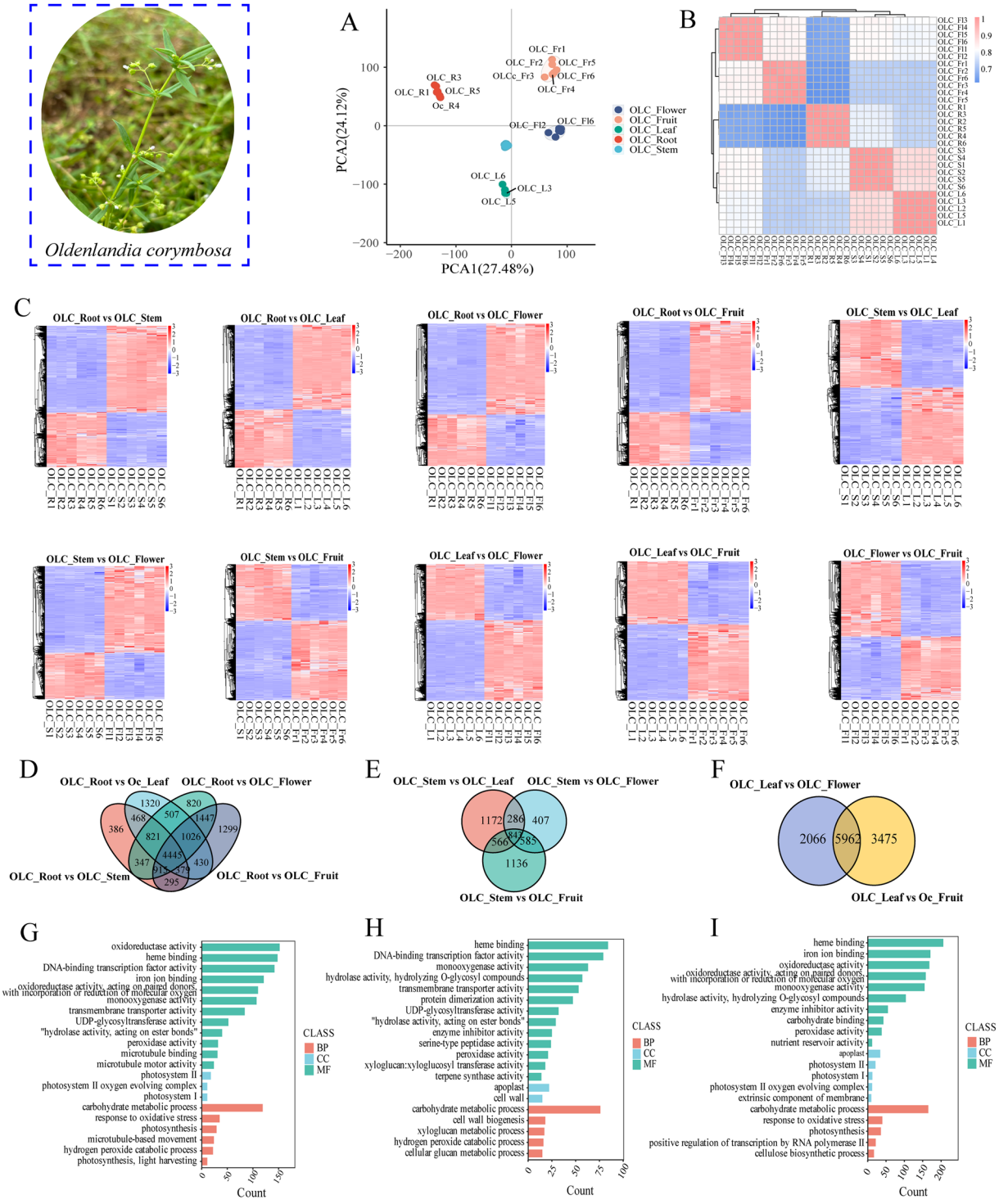


Fig. 5 Transcriptome data analysis of *Oldenlandia corymbose*, including expression profiles and DEG analysis for each comparative combination (**A**). Principal component analysis (PCA) plot of transcriptome samples (**B**). Heat map of gene associations between transcriptome samples (**C**). DEG expression pattern for each paired combination. Up-regulated DEGs are highlighted in red; down-regulated DEGs are shown in blue. DEG with padj < 0.05 and $|FC| > 2$ screening (**D**). OLC_Root vs OLC_Stem, OLC_Root vs OLC_Leaf, OLC_Root vs OLC_Flower, OLC_Root vs OLC_Fruit (**E**). OLC_Stem vs OLC_Leaf, OLC_Stem vs OLC_Flower, OLC_Stem vs OLC_Fruit, and (**F**). OLC_Leaf vs OLC_Flower and OLC_Leaf vs OLC_Fruit compare Venn diagrams sharing DEG between combinations (**G**). GO analysis of shared DEG between OLC_Root vs OLC_Stem, OLC_Root vs OLC_Leaf, OLC_Root vs OLC_Flower, OLC_Root vs OLC_Fruit comparison combinations (top 20 GO entries) (**H**). GO analysis of shared DEG between OLC_Stem vs OLC_Leaf, OLC_Stem vs OLC_Flower, OLC_Stem vs OLC_Fruit comparison combinations (top 20 GO entries) (**I**). GO analysis of shared DEG between OLC_Leaf vs OLC_Flower, and OLC_Leaf vs OLC_Fruit comparison combinations (top 20 GO entries). Note: OLC represents samples from *Oldenlandia corymbose*, R, S, L, Fl, Fr represent Roots, Stems, Leaves, Flowers and Fruits respectively. -1, -2, -3, -4, -5, -6 represent six biological repeat.

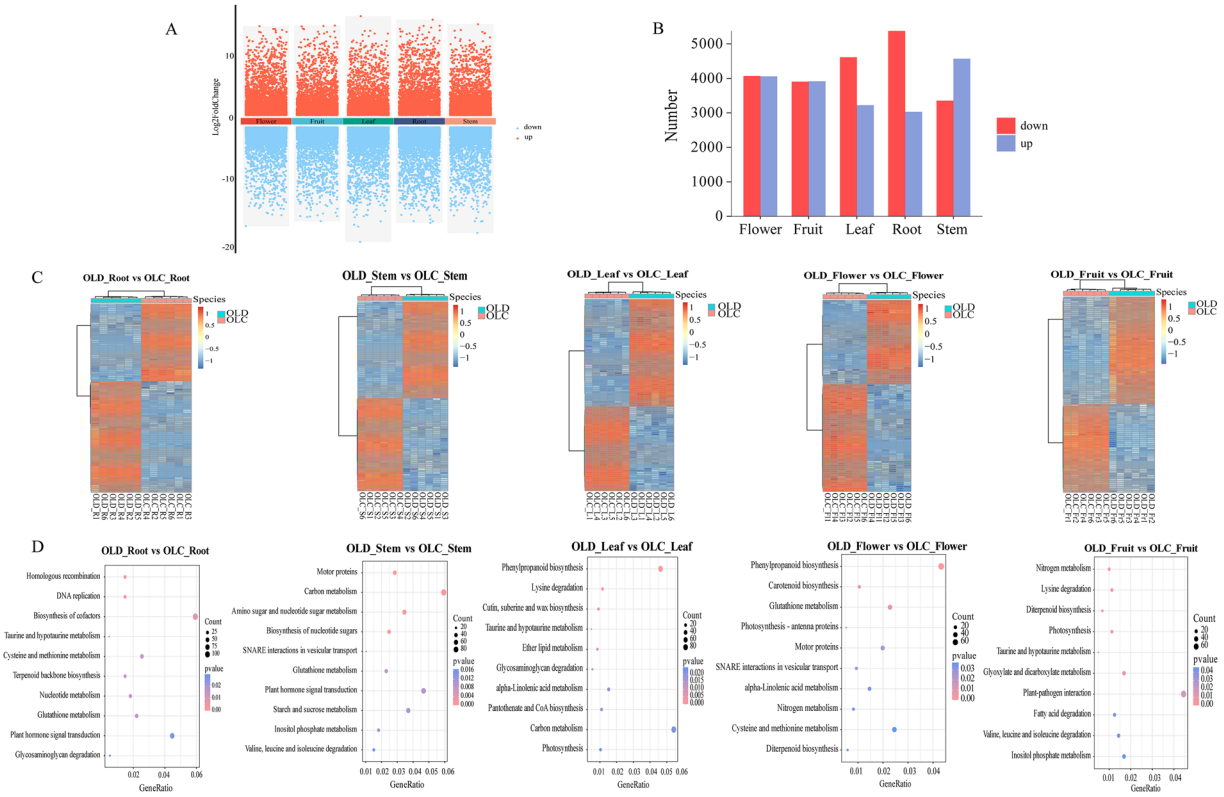


Fig. 6 Comparative transcriptome analysis of *Oldenlandia corymbose* and *Oldenlandia diffusa* (**A**). a multi-group volcano plot illustrating differentially expressed genes across Root, Stem, Leaf, Flower, and Fruit tissues of the two species, with OLD serving as the control group and OLC as the treatment group (**B**). Using OLD as the control group and OLC as the treatment group, a bar chart was generated to illustrate the differentially expressed genes in the Root, Stem, Leaf, Flower, and Fruit of the two species (**C**). Using OLD as the control group and OLC as the treatment group, a clustered heat map was constructed to visualize the top 100 differentially expressed genes across Root, Stem, Leaf, Flower, and Fruit tissues of the two species (**D**). Using OLD as the control group and OLC as the treatment group, KEGG enrichment analysis was conducted to identify functional pathways associated with differentially expressed genes in Root, Stem, Leaf, Flower, and Fruit tissues of the two species.

The third Venn diagram compared DEMs between leaf tissue (a vegetative organ) and reproductive organs (flower and fruit) (OLC_Leaf vs. OLC_Flower, OLC_Leaf vs. OLC_Fruit). In Oldenlandia diffusa, 287 shared DEMs were monitored (Fig. 3C), primarily related to biosynthesis of secondary metabolites, flavonoid biosynthesis, and anthocyanin biosynthesis (Fig. 3F). In contrast, *Oldenlandia corymbosa* displayed 325 shared DEMs (Fig. 3I), associated with flavonoid biosynthesis, anthocyanin biosynthesis, and 2-oxocarboxylic acid metabolism (Fig. 3L).

Identification of differentially expressed genes (DEGs) among different tissues of *Oldenlandia diffusa* and *Oldenlandia corymbosa*. To explore inter-tissue differences in *Oldenlandia diffusa* and *Oldenlandia corymbosa*, we analyzed 10 comparison combinations of tissue samples from both species: Root vs Stem, Root vs Leaf, Root vs Flower, Root vs Fruit, Stem vs Leaf, Stem vs Flower, Stem vs Fruit, Leaf vs Flower, Leaf vs Fruit, and Flower vs Fruit. These comparisons aimed to identify differentially expressed genes (DEGs) across tissue types. The results were visualized in heat maps (Figs. 4C, 5C).

Next, Venn diagrams were constructed to illustrate the shared differentially expressed genes (DEGs) in various comparison combinations (Fig. 4D–F, Fig. 5D–F). The first Venn diagram compared DEGs between roots and above-ground tissues (stem, leaf, flower, fruit) to identify DEGs between underground and above-ground tissues. In *Oldenlandia diffusa*, 8521 shared DEGs were found (Fig. 4D), which were primarily involved in monooxygenase activity, photosynthesis, DNA-binding transcription factor activity, transmembrane transport, and carbohydrate metabolic processes (Fig. 4G). In contrast, *Oldenlandia corymbosa* had 4445 shared DEGs (Fig. 5D) that participated in oxidoreductase activity, heme binding, photosynthesis, response to oxidative stress, and carbohydrate metabolic processes (Fig. 5G). This suggested that the functional differences of similar DEGs between above-ground and underground tissues in the two species were mainly reflected in photosynthesis and the synthesis regulation of various basic life substances. The second Venn diagram analyzed the shared DEGs between the stem and other above-ground tissues: Stem vs Leaf, Stem vs Flower, and Stem vs Fruit. In *Oldenlandia diffusa*, 3340 shared DEGs were found (Fig. 4E), which were involved in hydrolase activity (hydrolyzing O-glycosyl compounds), DNA-binding transcription factor activity, carbohydrate metabolic process, UDP-glycosyltransferase activity, and monooxygenase activity (Fig. 4H). In *Oldenlandia corymbosa*, 843 shared differentially expressed genes (DEGs)

Sample	Total reads	Total bases	Clean reads	Clean bases	Q30 rate (%)	GC (%)
OLD_R1	89,161,370	13,374,205,500	89,107,874	12,756,337,230	94.74	41.76
OLD_R2	65,971,550	9,895,732,500	65,932,790	9,850,514,486	92.64	44.52
OLD_R3	59,625,878	8,943,881,700	59,591,558	8,896,523,212	93.21	44.66
OLD_R4	58,998,370	8,849,755,500	58,963,998	8,811,128,814	92.37	44.4
OLD_R5	68,598,756	10,289,813,400	68,559,600	10,242,596,688	92.76	44.7
OLD_R6	50,261,690	7,539,253,500	50,233,250	7,508,530,554	92.82	44.36
OLD_S1	51,349,016	7,702,352,400	51,344,284	7,677,441,346	93.41	43.87
OLD_S2	58,139,628	8,720,944,200	58,105,430	8,681,408,544	92.58	43.75
OLD_S3	58,012,660	8,701,899,000	57,978,376	8,652,113,572	92.52	43.84
OLD_S4	60,894,794	9,134,219,100	60,857,522	9,031,134,028	93.95	45.05
OLD_S5	35,583,462	5,337,519,300	35,561,314	5,302,516,944	93.48	45.2
OLD_S6	50,203,754	7,530,563,100	50,173,542	7,466,745,600	93.16	45.94
OLD_L1	55,680,442	8,352,066,300	55,646,876	8,309,534,368	93.07	44.47
OLD_L2	70,927,262	10,639,089,300	70,883,230	10,567,267,298	94.02	43.93
OLD_L3	65,233,304	9,784,995,600	65,193,334	9,718,086,748	93.39	44.43
OLD_L4	57,214,496	8,582,174,400	57,180,448	8,535,391,710	92.88	44.56
OLD_L5	63,454,418	9,518,162,700	63,418,508	9,477,921,558	92.96	45.23
OLD_L6	63,091,060	9,463,659,000	63,053,164	9,405,909,330	92.96	44.12
OLD_Fl1	54,127,646	8,119,146,900	54,095,746	8,073,571,620	92.53	44.33
OLD_Fl2	58,881,626	8,832,243,900	58,849,426	8,787,155,302	92.83	43.43
OLD_Fl3	57,570,450	8,635,567,500	57,538,102	8,595,297,310	92.65	44.4
OLD_Fl4	95,835,932	14,375,389,800	95,722,126	13,960,706,688	96.3	43.36
OLD_Fl5	47,776,200	7,166,430,000	47,771,764	7,141,134,140	93.02	45.17
OLD_Fl6	51,886,420	7,782,963,000	51,855,550	7,755,249,354	92.89	44.63
OLD_Fr1	52,520,020	7,878,003,000	52,489,174	7,843,249,626	92.56	45.22
OLD_Fr2	49,452,752	7,417,912,800	49,423,942	7,376,752,918	92.38	45.2
OLD_Fr3	56,947,568	8,542,135,200	56,914,132	8,498,693,528	92.78	45.15
OLD_Fr4	64,760,376	9,714,056,400	64,722,940	9,665,449,566	92.81	44.91
OLD_Fr5	54,250,974	8,137,646,100	54,217,486	8,100,853,794	93.82	45.1
OLD_Fr6	104,329,444	15,649,416,600	104,223,352	15,432,761,858	94.15	46.23
OLC_R1	63,491,116	9,523,667,400	63,484,828	9,497,625,346	93.98	44.28
OLC_R2	62,375,870	9,356,380,500	62,338,812	9,306,431,986	92.86	44.37
OLC_R3	65,629,156	9,844,373,400	65,592,752	9,813,370,696	93.52	44.77
OLC_R4	57,894,336	8,684,150,400	57,858,222	8,567,292,388	93.91	46
OLC_R5	52,215,574	7,832,336,100	52,210,036	7,724,390,710	94.49	45.88
OLC_R6	43,960,436	6,594,065,400	43,933,814	6,536,592,044	93.16	46.53
OLC_S1	67,055,868	10,058,380,200	67,014,014	9,978,653,644	93.57	44.49
OLC_S2	77,837,572	11,675,635,800	77,788,444	11,571,371,108	93.98	44.1
OLC_S3	60,169,212	9,025,381,800	60,135,020	8,955,165,924	93.24	44.28
OLC_S4	52,292,944	7,843,941,600	52,263,220	7,808,016,526	92.31	44.33
OLC_S5	56,881,254	8,532,188,100	56,847,064	8,478,115,228	93.2	45.5
OLC_S6	59,417,122	8,912,568,300	59,383,386	8,863,713,088	92.73	46.05
OLC_L1	62,077,018	9,311,552,700	62,040,092	9,262,013,172	92.7	44.62
OLC_L2	47,025,620	7,053,843,000	46,998,590	7,026,477,806	92.27	44.07
OLC_L3	55,455,428	8,318,314,200	55,422,828	8,284,296,532	92.58	44.19
OLC_L4	57,630,668	8,644,600,200	57,596,146	8,610,747,832	92.45	44.35
OLC_L5	55,671,234	8,350,685,100	55,638,882	8,312,530,150	92.39	44.08
OLC_L6	58,141,106	8,721,165,900	58,107,796	8,675,853,656	92.57	44.38
OLC_Fl1	53,455,190	8,018,278,500	53,424,260	7,976,233,840	92.59	44.72
OLC_Fl2	48,349,492	7,252,423,800	48,323,058	7,221,485,542	91.36	45.01
OLC_Fl3	50,608,800	7,591,320,000	50,580,636	7,546,429,588	92.81	43.84
OLC_Fl4	40,724,226	6,108,633,900	40,700,322	6,036,934,602	93.35	46.39
OLC_Fl5	59,393,828	8,909,074,200	59,358,376	8,855,821,730	92.96	44.24
OLC_Fl6	60,548,442	9,082,266,300	60,514,174	9,034,180,426	92.89	44.34
OLC_Fr1	74,521,046	11,178,156,900	74,476,866	11,088,312,214	92.89	44.61
OLC_Fr2	71,489,154	10,723,373,100	71,445,042	10,607,474,374	93.88	44.5
OLC_Fr3	54,623,968	8,193,595,200	54,589,880	8,129,798,536	93.89	46.09

Continued

Sample	Total reads	Total bases	Clean reads	Clean bases	Q30 rate (%)	GC (%)
OLC_Fr4	58,007,340	8,701,101,000	41,859,986	6,214,041,396	93.15	45.56
OLC_Fr5	54,756,438	8,213,465,700	54,725,124	8,179,679,074	92.18	44.85
OLC_Fr6	59,500,932	8,925,139,800	59,465,314	8,876,214,432	92.79	45.08

Table 1. Transcriptome Clean data production statistics. **Note:** OLC and OLD represent samples from *Oldenlandia diffusa* and *Oldenlandia corymbosa* respectively. R, S, L, Fl, and Fr represent roots, stems, leaves, flowers, and fruits respectively. -1, -2, -3, -4, -5, -6 represent six biological replicates.

were identified (Fig. 5E). These shared DEGs were associated with various biological processes, including hydrolase activity, hydrolyzing O - glycosyl compounds, heme binding, DNA-binding transcription factor activity, and monooxygenase activity (Fig. 5H). The third Venn diagram compared the DEGs between the leaf (a vegetative organ) and the reproductive organs (flower and fruit) (OLC_Leaf vs OLC_Flower, OLC_Leaf vs OLC_Fruit). In *Oldenlandia diffusa*, 6,453 shared DEGs were identified (Fig. 4F). These shared DEGs were involved in processes such as monooxygenase activity, DNA-binding transcription factor activity, UDP-glycosyltransferase activity, and oxidoreductase activity (Fig. 4I). In contrast, 5,962 shared DEGs were obtained in *Oldenlandia corymbosa* (Fig. 5F). The shared DEGs were associated with heme binding, iron ion binding, monooxygenase activity, oxidoreductase activity, acting on paired donors, with incorporation or reduction of molecular oxygen (Fig. 5I).

Identification of differentially expressed genes (DEGs) between *Oldenlandia diffusa* and *Oldenlandia corymbosa*. For a comparative analysis of the same tissue types between the two species, sequencing reads were independently aligned to the respective reference genome annotations, and gene-level read counts were generated. To ensure cross-species comparability, reciprocal BLAST was performed using TBtools, with homologous gene pairs identified under the following criteria: e-value $\leq 1 \times 10^{-5}$ and bidirectional coverage $\geq$70%. Only one-to-one orthologous gene pairs were retained for downstream analysis. Following count matrix alignment, differential expression analysis was conducted using edgeR. A log-transformed "exon union length" was incorporated as a gene-specific offset to account for transcript length bias. For orthology relationships involving one-to-many or many-to-one mappings, orthogonal groups were resolved based on bidirectional best hits with connected components analysis. In such cases, gene counts were aggregated at the orthogroup level per species, and edgeR modeling with length correction was applied accordingly. All analyses were implemented using custom R scripts. Figure 6 presents the differential expression analysis of orthologous genes in one-to-one ortholog sets. Compared with OLD, OLC exhibited up-regulation of 3,028, 4,569, 3,222, 4,055, and 3,914 genes in Root, Stem, Leaf, Flower, and Fruit, respectively, while down-regulated genes numbered 5,372, 3,353, 461, 4,608, and 3,903 (Fig. 6A,B). Additionally, we examined the differential expression profiles of the top 100 most highly expressed genes in each tissue across the two species (Fig. 6C). Furthermore, orthologous gene pairs with consistent functional annotations were selected for KEGG pathway enrichment analysis. In roots, differentially expressed genes were predominantly enriched in pathways including DNA replication, homologous recombination, cofactor biosynthesis, and terpenoid backbone biosynthesis. In stems, significant enrichments were observed in motor proteins, carbon metabolism, amino sugar and nucleotide sugar metabolism, and nucleotide sugar biosynthesis. In leaves, enriched pathways included phenylpropanoid biosynthesis, lysine degradation, and Cutin, suberine, and wax biosynthesis. In flowers, enriched pathways encompassed Phenylpropanoid biosynthesis, Carotenoid biosynthesis, and Glutathione metabolism. In fruits, differentially expressed genes were mainly associated with Nitrogen metabolism, Lysine degradation, and Diterpenoid biosynthesis (Fig. 6D).

Data Records

In this study, raw transcriptome sequencing data are available in the National Center for Biotechnology Information (NCBI, https://identifiers.org/ncbi/insdc.sra:SRP665145) under Accession PRJNA1405479 and SRP665145[27]. The data include 60 samples of *Oldenlandia diffusa*(OLD) and *Oldenlandia corymbose*(OLC), numbered CRR1765877 to CRR1765930. The original metabolome data was uploaded to MetaboLights(MTBLS13314, https://www.ebi.ac.uk/metabolights/MTBLS13314)[28]. Additionally, expression and quality control data for metabolome and transcriptome analyses have been uploaded to the Figshare database for accessibility[29].

Technical Validation

Validation of metabolomic data. The metabolomics data were collected from 60 samples across two species. Mass spectrometry data were processed using Analyst 1.6.3 software. Qualitative and quantitative analyses of metabolites were conducted based on a local metabolite database, with MultiQuant software used for chromatographic peak integration and correction. Chromatographic peaks of each detected metabolite were corrected according to retention time and peak shape information to ensure the accuracy of the analyses. In total, 1,342 metabolites were detected using the UPLC-MS/MS platform and the self-built database. Principal component analysis (PCA) of the metabolites revealed significant separation among samples from different tissues of the two species, with the first three principal components (PC1, PC2, and PC3) accounting for 24.67%, 15.85%, and 12.77% of the variation, respectively (Fig. 2A). The quality control (QC) samples exhibited high data stability, with over 85% of the substances having a coefficient of variation (CV) less than 0.5 and over 75% having a CV less than 0.3. Meanwhile, the test samples in each group displayed a CV curve aligned with that of the QC samples (Fig. 2B). Additionally, the six QC samples demonstrated high repeatability (Fig. 2C), and there were strong correlations among the six biological replicates of the same tissue, confirming the reliability of the

Sample name	Total reads	Total mapped	Multiple mapped	Uniquely mapped
OLD_R1	89107874	84527733 (94.86%)	1512458 (1.70%)	83015275 (93.16%)
OLD_R2	65932790	62723373 (95.13%)	1207458 (1.83%)	61515915 (93.30%)
OLD_R3	59591558	56742538 (95.22%)	1061067 (1.78%)	55681471 (93.44%)
OLD_R4	58963998	56231116 (95.37%)	1063822 (1.80%)	55167294 (93.56%)
OLD_R5	68559600	65122301 (94.99%)	1217733 (1.78%)	63904568 (93.21%)
OLD_R6	50233250	47699978 (94.96%)	887587 (1.77%)	46812391 (93.19%)
OLD_S1	51344284	50016600 (97.41%)	890919 (1.74%)	49125681 (95.68%)
OLD_S2	58105430	56223113 (96.76%)	1009088 (1.74%)	55214025 (95.02%)
OLD_S3	57978376	55994694 (96.58%)	987553 (1.70%)	55007141 (94.88%)
OLD_S4	60857522	58896038 (96.78%)	1064072 (1.75%)	57831966 (95.03%)
OLD_S5	35561314	34465694 (96.92%)	1133439 (3.19%)	33332255 (93.73%)
OLD_S6	50173542	48601862 (96.87%)	1272408 (2.54%)	47329454 (94.33%)
OLD_L1	55646876	53806327 (96.69%)	1058250 (1.90%)	52748077 (94.79%)
OLD_L2	70883230	66873760 (94.34%)	1250125 (1.76%)	65623635 (92.58%)
OLD_L3	65193334	62971492 (96.59%)	1223954 (1.88%)	61747538 (94.71%)
OLD_L4	57180448	55209886 (96.55%)	1127999 (1.97%)	54081887 (94.58%)
OLD_L5	63418508	61306633 (96.67%)	1357493 (2.14%)	59949140 (94.53%)
OLD_L6	63053164	60837768 (96.49%)	1374874 (2.18%)	59462894 (94.31%)
OLD_Fl1	54095746	50302082 (92.99%)	1167769 (2.16%)	49134313 (90.83%)
OLD_Fl2	58849426	54998127 (93.46%)	946823 (1.61%)	54051304 (91.85%)
OLD_Fl3	57538102	53206304 (92.47%)	1085665 (1.89%)	52120639 (90.58%)
OLD_Fl4	95722126	87529621 (91.44%)	2281478 (2.38%)	85248143 (89.06%)
OLD_Fl5	47771764	44261156 (92.65%)	924697 (1.94%)	43336459 (90.72%)
OLD_Fl6	51855550	48809623 (94.13%)	936267 (1.81%)	47873356 (92.32%)
OLD_Fr1	52489174	50575086 (96.35%)	1402724 (2.67%)	49172362 (93.68%)
OLD_Fr2	49423942	47740202 (96.59%)	1742499 (3.53%)	45997703 (93.07%)
OLD_Fr3	56914132	55413040 (97.36%)	1382451 (2.43%)	54030589 (94.93%)
OLD_Fr4	64722940	63004784 (97.35%)	1696026 (2.62%)	61308758 (94.72%)
OLD_Fr5	54217486	52930771 (97.63%)	1441467 (2.66%)	51489304 (94.97%)
OLD_Fr6	104223352	98135497 (94.16%)	10200277 (9.79%)	87935220 (84.37%)
OLC_R1	63484828	57220962 (90.13%)	1050397 (1.65%)	56170565 (88.48%)
OLC_R2	62338812	55419376 (88.90%)	1616038 (2.59%)	53803338 (86.31%)
OLC_R3	65592752	58879763 (89.77%)	2816445 (4.29%)	56063318 (85.47%)
OLC_R4	57858222	50351271 (87.03%)	2088699 (3.61%)	48262572 (83.42%)
OLC_R5	52210036	46854460 (89.74%)	1586121 (3.04%)	45268339 (86.70%)
OLC_R6	43933814	39143193 (89.10%)	4406500 (10.03%)	34736693 (79.07%)
OLC_S1	67014014	54242731 (80.94%)	1246510 (1.86%)	52996221 (79.08%)
OLC_S2	77788444	62357798 (80.16%)	1183039 (1.52%)	61174759 (78.64%)
OLC_S3	60135020	48571459 (80.77%)	795968 (1.32%)	47775491 (79.45%)
OLC_S4	52263220	42204885 (80.75%)	1072076 (2.05%)	41132809 (78.70%)
OLC_S5	56847064	46246907 (81.35%)	924184 (1.63%)	45322723 (79.73%)
OLC_S6	59383386	47977003 (80.79%)	1600115 (2.69%)	46376888 (78.10%)
OLC_L1	62040092	59578127 (96.03%)	1559105 (2.51%)	58019022 (93.52%)
OLC_L2	46998590	44951140 (95.64%)	891817 (1.90%)	44059323 (93.75%)
OLC_L3	55422828	52985839 (95.60%)	1122077 (2.02%)	51863762 (93.58%)
OLC_L4	57596146	55257206 (95.94%)	1148096 (1.99%)	54109110 (93.95%)
OLC_L5	55638882	53432669 (96.03%)	1030089 (1.85%)	52402580 (94.18%)
OLC_L6	58107796	55540350 (95.58%)	1693825 (2.91%)	53846525 (92.67%)
OLC_Fl1	53424260	49437244 (92.54%)	1658292 (3.10%)	47778952 (89.43%)
OLC_Fl2	48323058	44971064 (93.06%)	2717473 (5.62%)	42253591 (87.44%)
OLC_Fl3	50580636	47126392 (93.17%)	789217 (1.56%)	46337175 (91.61%)
OLC_Fl4	40700322	38277292 (94.05%)	765224 (1.88%)	37512068 (92.17%)
OLC_Fl5	59358376	55586756 (93.65%)	1068211 (1.80%)	54518545 (91.85%)
OLC_Fl6	60514174	56808816 (93.88%)	2074230 (3.43%)	54734586 (90.45%)
OLC_Fr1	74476866	72660330 (97.56%)	1817489 (2.44%)	70842841 (95.12%)
OLC_Fr2	71445042	68959997 (96.52%)	2789702 (3.90%)	66170295 (92.62%)
OLC_Fr3	54589880	52980442 (97.05%)	1678093 (3.07%)	51302349 (93.98%)

Continued

Sample name	Total reads	Total mapped	Multiple mapped	Uniquely mapped
OLC_Fr4	41859986	40550612 (96.87%)	2840673 (6.79%)	37709939 (90.09%)
OLC_Fr5	54725124	53283861 (97.37%)	1233114 (2.25%)	52050747 (95.11%)
OLC_Fr6	59465314	58007258 (97.55%)	1373958 (2.31%)	56633300 (95.24%)

Table 2. Statistical table of comparison rate between each sample and reference genome in *Oldenlandia diffusa* and *Oldenlandia corymbosa*. **Note:** OLC and OLD represent samples from *Oldenlandia diffusa* and *Oldenlandia corymbosa* respectively. R, S, L, Fl, and Fr represent roots, stems, leaves, flowers, and fruits respectively. -1, -2, -3, -4, -5, -6 represent six biological replicates.

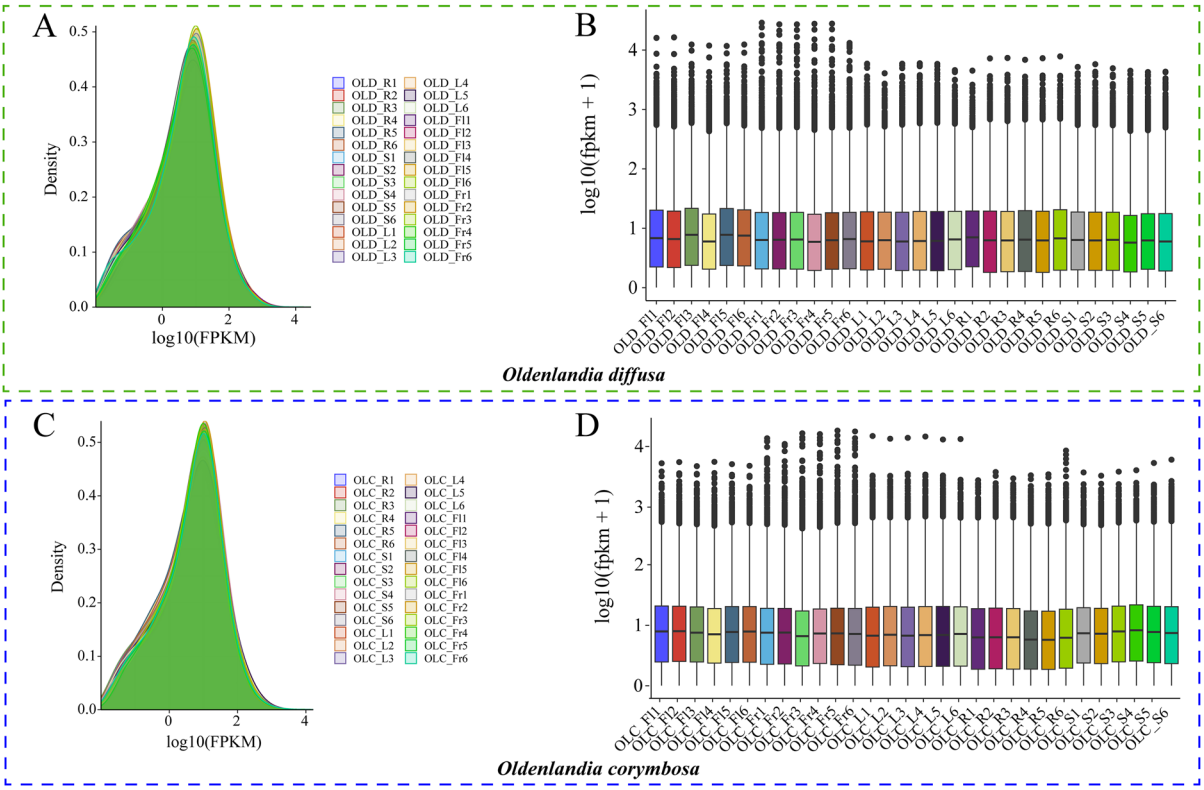


Fig. 7 Quality control chart of transcriptome data for 60 samples by *Oldenlandia diffusa* and *Oldenlandia corymbosa* (**A,C**). Comparison of FPKM density distribution (**B,D**). Box plots of expression levels of different samples (fpkm + 1).

measurements (Fig. 2E). The detection process resulted in a comprehensive metabolomics dataset comprising Flavonoids (24.96%), Phenolic acids (21.09%), Alkaloids (14.38%), Terpenoids (12.22%), among others (Fig. 2D).

Validation of transcriptome data. We obtained 272.121 Gb and 257.039 Gb of clean data from samples of *Oldenlandia diffusa* and *Oldenlandia corymbosa*, respectively. The Q30 ratio in each sample exceeded 91.36%, and the GC content was no less than 41.76% (Table 1). For the *Oldenlandia diffusa* samples, the total mapped rate of sequences to their reference genome ranged from 91.44% to 97.63%, the multiple mapped rate ranged from 1.61% to 9.79%, and the uniquely mapped rate ranged from 84.37% to 95.03%. For the *Oldenlandia corymbosa* samples, the total mapped rate of sequences to their reference genome ranged from 80.75% to 97.56%, the multiple mapped rate ranged from 1.32% to 6.79%, and the uniquely mapped rate ranged from 78.10% to 95.24% (Table 2). Principal component analysis revealed distinct clustering of tissue samples, with PC1 and PC2 explaining 28.65% and 19.84% of the variance in *Oldenlandia diffusa*, and 27.48% and 24.12% in *Oldenlandia corymbosa* (Figs. 3A, 4A). Samples from the same tissue exhibited strong correlations, and a certain correlation was generated between the flower and fruit samples of *Oldenlandia diffusa* (Figs. 3B, 4B). We obtained 37,644 and 20,825 annotated genes from *Oldenlandia diffusa* and *Oldenlandia corymbosa*, respectively. Analysis of the FPKM density plot revealed that most gene expression levels clustered between 1 and 10, with coding gene FPKM values spanning from 10^{-2} to 10^{4} (Fig. 7A,C). Box plots illustrated the distribution of gene expression levels across samples from the two species. The overall expression levels, quantified as $\log_{10}$ (FPKM + 1), ranged from 0 to 4.5. The high consistency among biological replicates within each group further validated the reliability of the transcriptome data (Fig. 7B,D).

Data availability

The original data of the present study cohort were deposited in the NCBI public repository (https://identifiers.org/ncbi/insdc.sra:SRP665145). Metabolomic raw data have been archived in MetaboLights (MTBLS13314, https://www.ebi.ac.uk/metabolights/MTBLS13314). Additionally, expression and quality control data for metabolome and transcriptome analyses have been uploaded to the figshare database for accessibility.

Code availability

The methods section details the software and R packages employed for metabolomic and transcriptomic analyses. The R scripts for the comparative transcriptome analysis of the two species have been stored in figshare.

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Author contributions

Pengyu Chen: Select topic, research proposal design, data analysis and paper writing. Zhuang Huang: Select topic and data analysis. Ping Huang: data analysis. Kaojiang Zhu, Benjiang Xiao and Meng Chen: data sorting. Yuxin Wen and Qi Jiang: Data organization and visualization. Rui Qian, Xing Hong, and Shihao Li: Data review and manuscript review. Fang Huang and Lintao Han: review and editing, experimental funding supporting.

Competing interests

The authors declare no competing interests.

Additional information

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